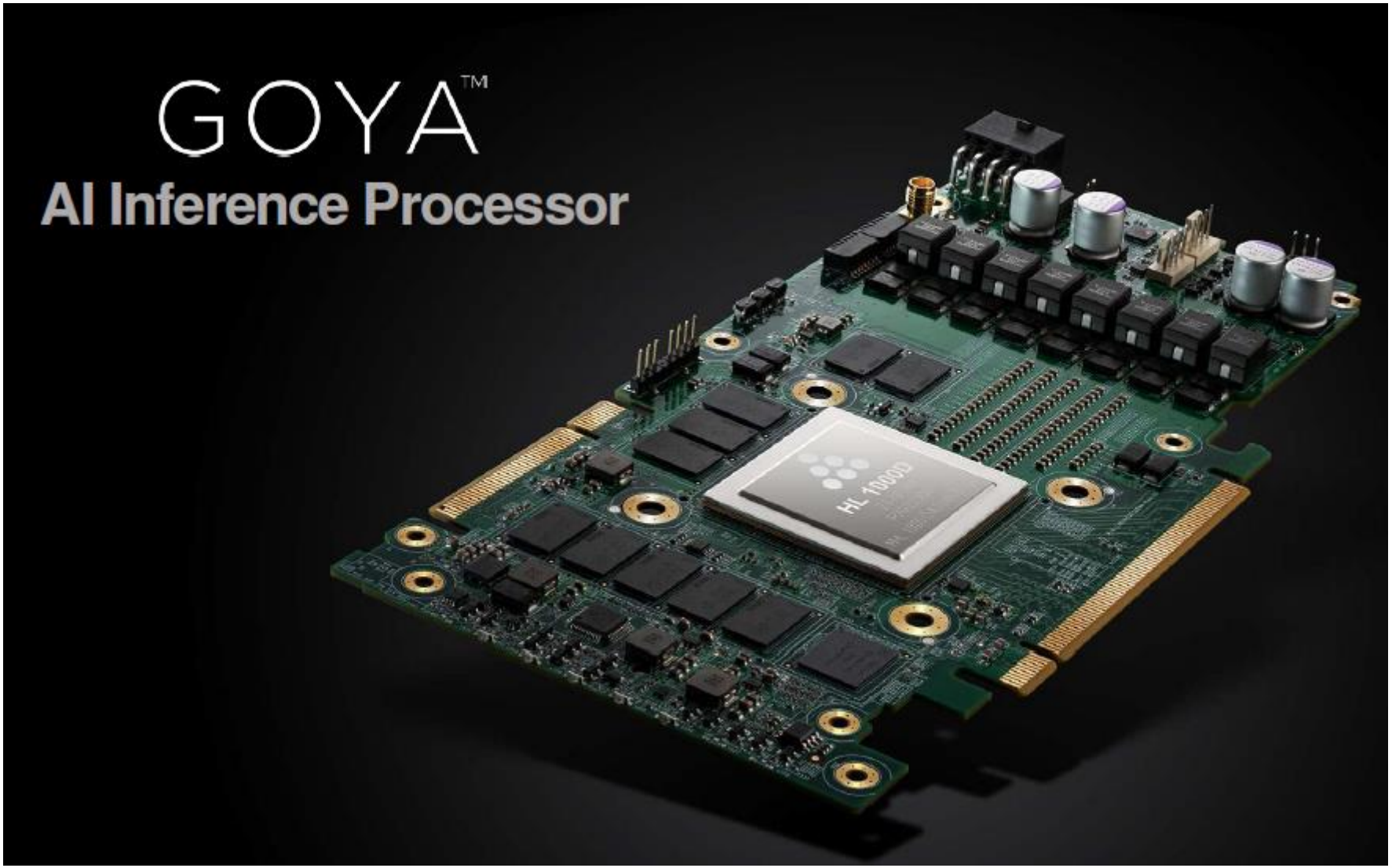
Intel Habana's Goya: An Inference Processor

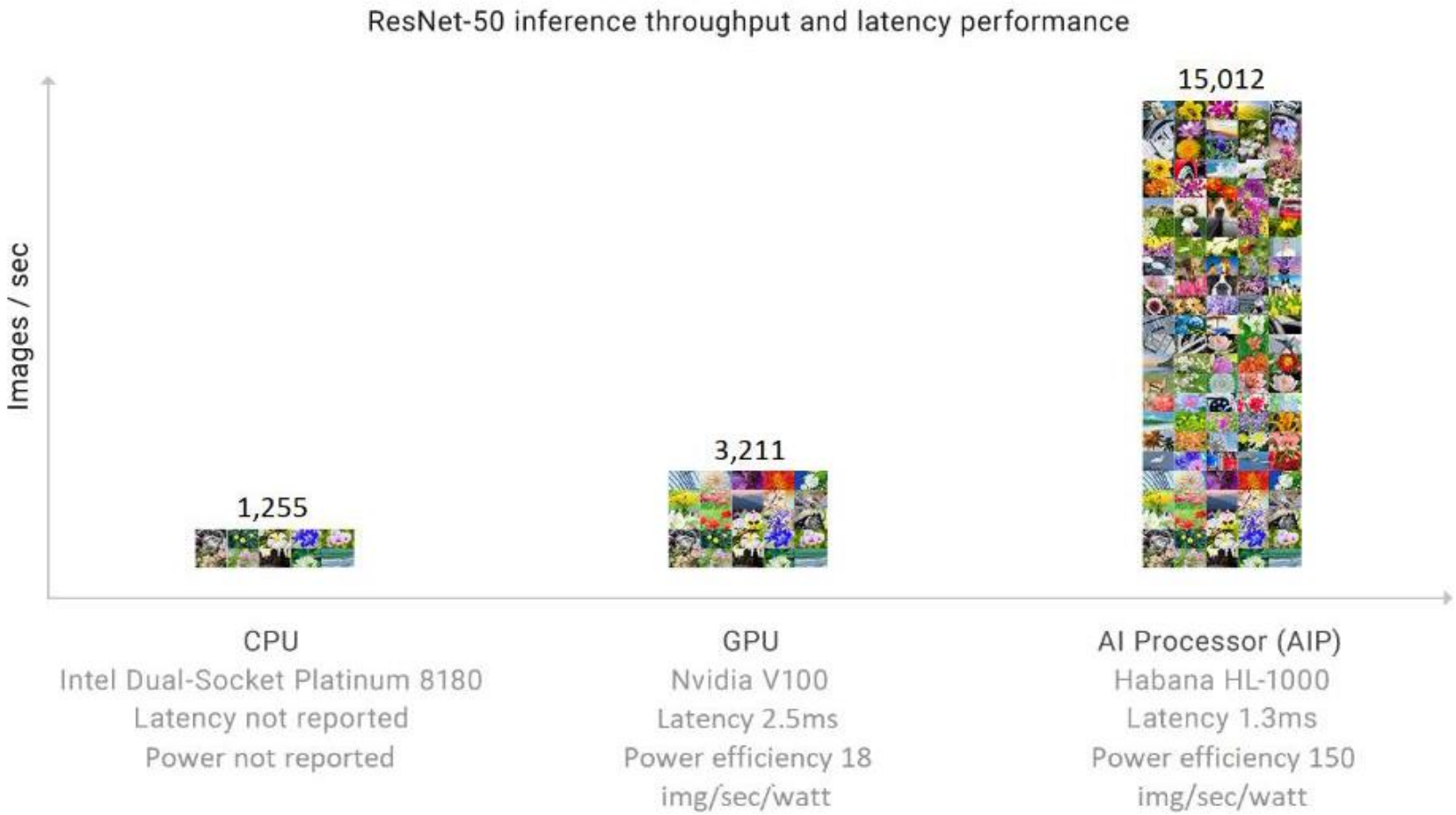
Sparsh Mittal

Acronyms

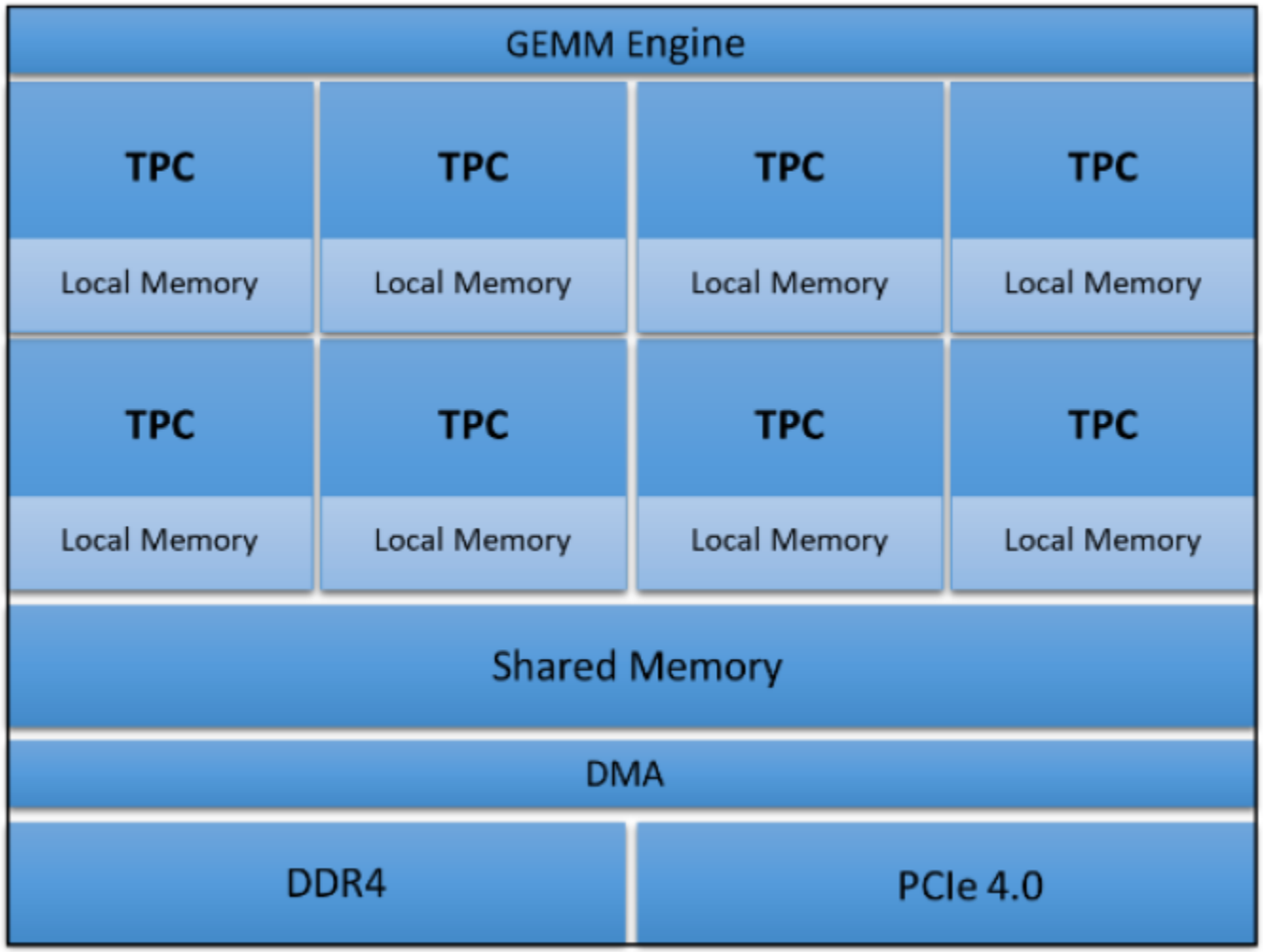
- VLIW: very long instruction word
- SIMD: single instruction multiple data


GOYA™
AI Inference Processor
HL 1000D

Appetizer!


ResNet-50 inference throughput and latency performance
Images / sec
1,255
3,211
15,012
CPU
Intel Dual-Socket Platinum 8180
Latency not reported
Power not reported
GPU
Nvidia V100
Latency 2.5ms
Power efficiency 18
img/sec/watt
AI Processor (AIP)
Habana HL-1000
Latency 1.3ms
Power efficiency 150
img/sec/watt

Goya architecture



Heterogeneous computing approach

- Goya has TPC and GEMM engine for accelerating different types of workloads. Similarly,
  - IBM's Power 10 CPU has a Matrix co-processor
  - Intel's Sapphire Rapids has a Tile Matrix Multiply (TMUL) unit
  - NVIDIA Volta GPU has CUDA and tensor cores
  - Intel's Xeon Phi has scalar and vector computing engines
  - Groq's TSP has vector and matrix units

Goya architecture

- Goya has fully programmable Tensor Processing Cores (TPCs) and GEMM engine
- TPC is a VLIW SIMD vector processor
- Features:
  - GEMM operation acceleration
  - Special functions dedicated hardware
  - Tensor addressing
  - Latency hiding capabilities
- Datatypes supported: FP32 • INT32 • INT16 • INT8 • UINT32 • UINT16 • UINT8
- SynapseAI quantizer selects suitable data type by balancing throughput and performance versus accuracy

Architecture

- GEMM engine operates on 16-bit integers
- TPC supports mixed-precision operations including 8-bit, 16-bit, and 32-bit SIMD vector operations for both integer and floating-point.
- Local memory is software-managed and not a hardware-managed cache
  - TPU, GPU, IPU etc all have software-managed memory
- On-die caches and memory can be either hardware-managed or fully software-managed, allowing the compiler to optimize the residency of data and reducing movement.

Architecture

- Each TPC has its own local memory and can also access SRAM-based shared memory.
- DMA engine acts as the intermediary between the shared memory and both external memory and I/O.
- GEMM, TPC, and DMA engines can operate concurrently with the shared memory, offering a latency-hiding mechanism.

Architectural parameters

- For predictability and low latency, Goya has
  - software-managed, on-die memory and
  - programmable centralized DMAs.
- For robustness, all memories are ECC-protected
- Goya PCIe card comes in single-slot or dual-slot form factors.
- 2 DDR4 channels @ 2.66 GT/s, 40 GB/s bandwidth and 4GB/8GB/16GB capacity
- Power required is not expected to exceed 100 to 140 watts for most inferencing work, but the card can draw up to 200 watts.

Limitations of Goya

- Slow DDR4 memory
- Relatively small SRAM capacity (TPUv4 has 144MB)
- No support for Bfloat16

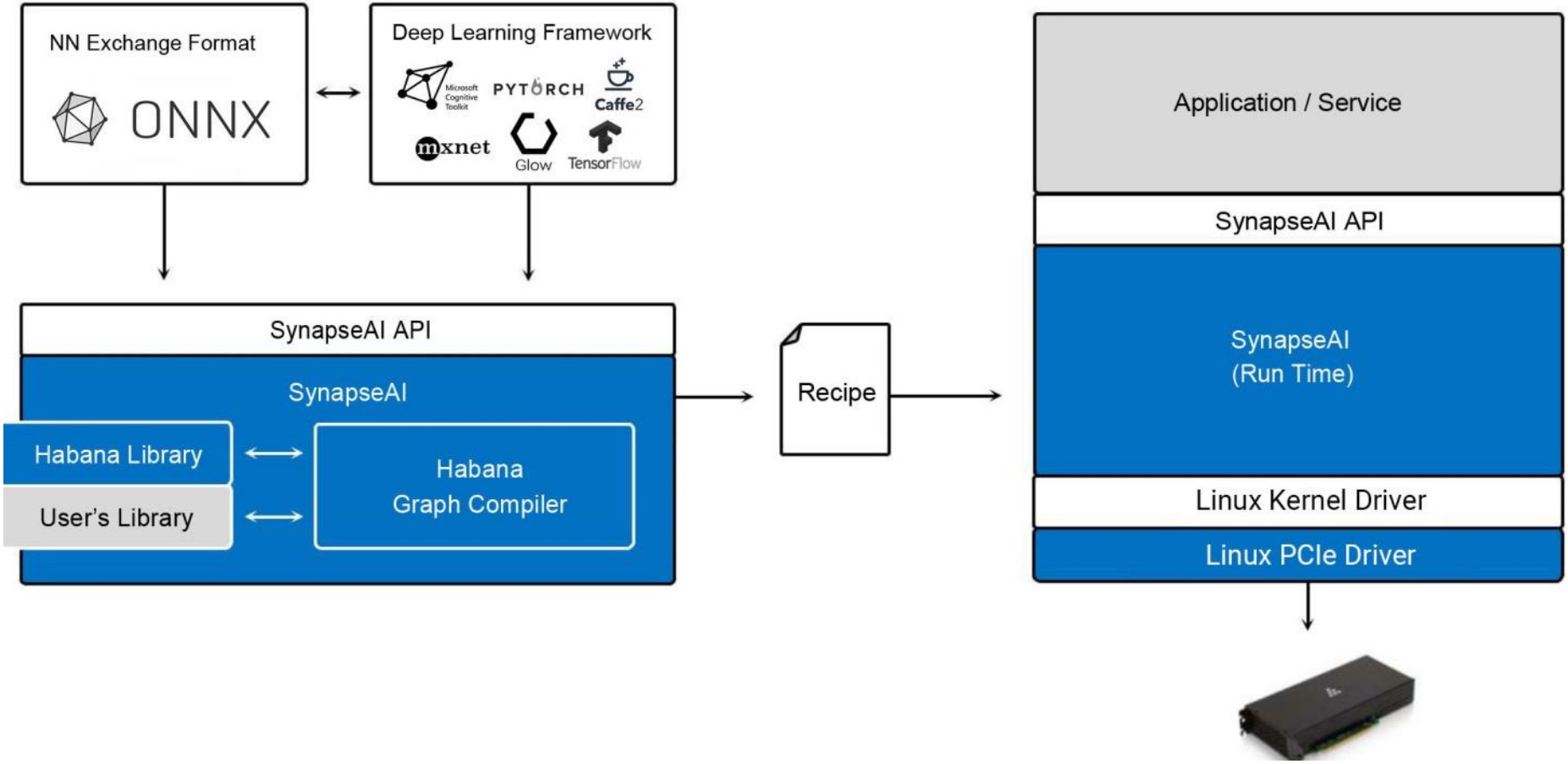
Performance is relative to T4 for MLPerf Inference 0.5

	perf/TDP vs T4	*TDP (W)*	Cores	*GHz*	*MiB SRAM*	*Type DRAM*
T4	1.00	70	40	0.6	20	GDDR6
Goya	0.46	200	8	2.1	≈32	DDR4
Nervana	0.69	100	24	1.1	60	LPDDR4
Zebra	0.25	225	--	--	54	DDR4
HanGuang	2.17	280	4	0.7	192	none

T4 is in 12nm, Goya is in 16nm

Jouppi et al., “Ten Lessons From Three Generations Shaped Google’s TPUv4i”, ISCA 2021

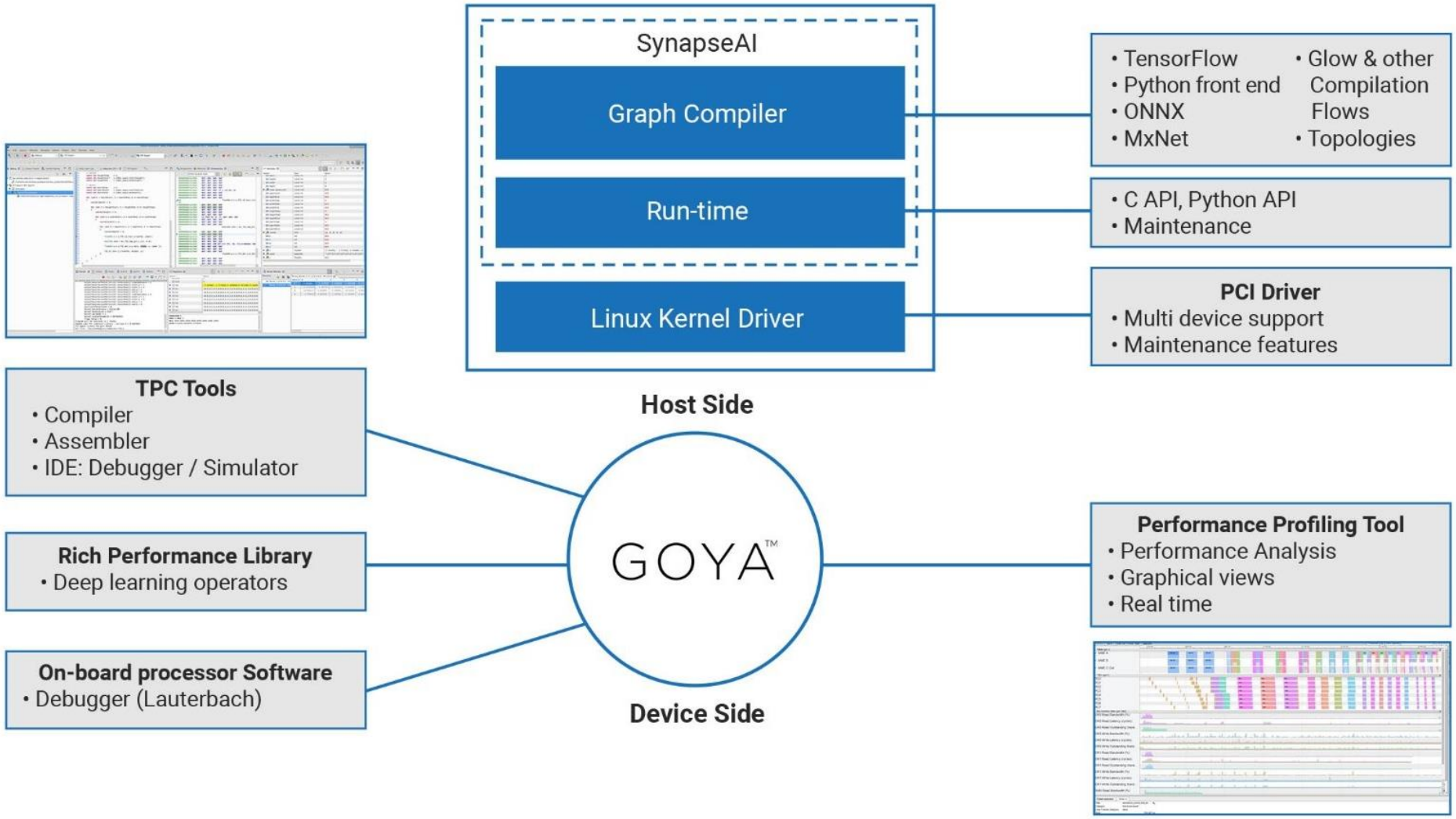
Software Framework



SynapseAI: an inference software toolkit

- It provides inference network model compilation (Graph Complier) and runtime.
- SynapseAI imports the trained model and compiles it for use on the Goya platform
- SynapseAI supports automatic quantization of models trained in floating-point format with near-zero accuracy loss.
- SynapseAI provides two APIs: C API and Python API
- Supports TensorFlow, MXNet, Caffe2, PyTorch, ONNX, CNTK and Glow compiler.
- ➔ Goya can use models trained on GPUs, CPUs or even TPUs.

Software development tools



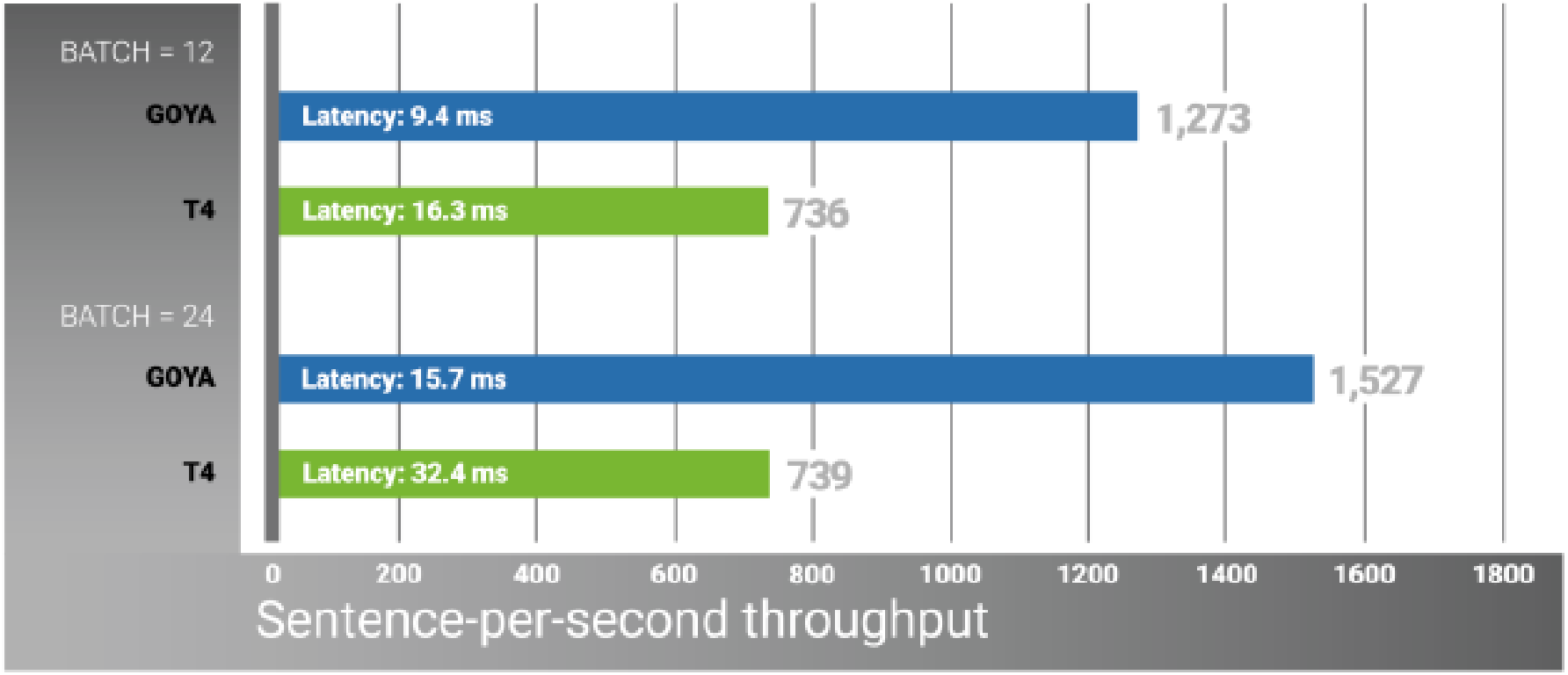
Experimental results

- Goya is implemented on 16nm
- ➔ Further improvements are possible by moving to 7nm or 3nm

Inference results

Topology	Frame Work	Perf Metric	Batch Size	Throughput	Latency (msec)
BERT SQuAD Base, Intermediate Size=3,072; Max Seq Len = 128	MXnet	Sentences/Sec	8	1108	7.2
			12	1273	9.4
			24	1527	15.7
Resnet18 v1	ONNX	Images/Sec	1	13156	0.1
			4	26384	0.3
			8	29961	0.5
			10	30726	0.6
Resnet34 v1	ONNX	Images/Sec	1	8345	0.2
			4	15050	0.5
			8	17254	0.8
			10	18016	0.9
Resnet50 v1	ONNX	Images/Sec	1	7466	0.2
			4	13221	0.5
			8	14546	0.9
			10	15453	1.0
Resnet101 v1	ONNX	Images/Sec	1	4533	0.4
			4	8062	0.9
			8	9086	1.3
			10	9705	1.5
Resnet152 v1	ONNX	Images/Sec	1	2912	0.6
			4	5138	1.2
			8	5737	1.9
			10	6075	2.2

BERT- Base SQuAD Inference Benchmark



Comparison with GPU/FPGA/CPU

	Goya	V100	Xilinx Alveo U250	Dual-socket Xeon (Skylake) Platinum 8180 CPU
Throughput (images/sec)	15012	3,247	3700	1225
Latency (ms)	1.3	2.5		
Energy efficiency (image/sec/watt)	150	18		6

For latency, lower is better. For other two, higher is better